Crude Oil Futures Hedging Strategy Based on Linear Regression

Project report

Abstract: hedge crude-oil and refined product futures using pairwise linear regression, beta-based hedge ratios, and portfolio-variance reduction.

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What this deck covers

- Asset universe and project setup
- Data organization and cleaning
- Linear regression, β, and R^2 ratio
- Minimum-variance hedging logic
- Dashboard design and outputs

1. Asset universe and project setup

Assets used

Ticker	Contract	Role in hedge system
Brent	ICE Brent crude	Global crude benchmark
WTI	NYMEX WTI crude	US crude benchmark
Heating Oil (HO)	NYMEX heating oil	Middle-distillate product proxy
Gas Oil (GO)	ICE gas oil	European distillate benchmark
RBOB	NYMEX gasoline blendstock	Gasoline product benchmark

Data preparation

- Collected 3 years of daily futures data from 28 Nov 2022 to 27 Nov 2025.
- Keep only common trading dates across all contracts to ensure consistency and comparability
- Closing prices were extracted and merged into a single price matrix for return and correlation computation

Code modules

- data_loader.py: import raw market data
- data_cleaner.py: align common dates
- regression.py: returns, correlation, pairwise OLS, beta and R^2 matrix
- portfolio_var.py: portfolio return and variance
- hedging.py: score and rank candidate hedges
- dashboard.py: Dash interface and interactive plots

2. Methodology: linear regression and hedge ratio

Regression model

$$r_y = \alpha + \beta r_x + \varepsilon$$

Where:

- r_y: return of the exposure asset
- r_x: return of the hedge asset
- α: intercept (average unexplained shift)
- β: sensitivity of the exposure asset to the hedge asset
- ε: residual risk not captured by the linear relation

Pairwise regression design in the code

- Generates all ordered pairs of assets using permutations, so each asset is tested as both hedge asset x and exposure asset y
- Estimates the gradient β, interpreted as the hedge ratio or sensitivity of the exposure asset to the hedge asset
- Compute R^2 to measure how well the hedge asset explains the variation to the exposure asset
- Stores results into a dataframe sorted by R^2 in descending order to identify asset pairs with best hedge fit

3. Hedging logic: maximise variance reduction per trading cost

Brent Heding Quantity

$$q_{\text{Brent}} = \begin{cases} -\text{Q}_\text{e}\,, if\ exposure\ asset\ is\ Brent \\ -\beta_{\text{e,Brent}}\text{Q}_\text{e}\,, otherwise \end{cases}$$

$$q_j = q_{Brent} \cdot \beta_{Brent,j}$$

Where:

- q_{Brent}: hedge quantity of Brent
- Q_e: quantity of exposure asset
- $\beta_{e,Brent}$: sensitivity of exposure asset to Brent
- q_j: hedge quantity of candidate assets
- $\beta_{e,Brent}$: sensitivity of Brent to candidate asset j

Logic:

- Use **Brent as the benchmark hedge route** to determine the base hedge quantity
- For each candidate hedge asset j, map the Brent hedge into that asset using the regression β

Hedging Score

$$\Delta\text{Var} = \text{Var}_{after} - \text{Var}_{before}$$

$$\text{Score} = -\frac{\Delta\text{Var}}{\text{Cost}}$$

Where:

- ΔVar measures the change in portfolio variance after hedging
- Higher score means more variance reduction per unit trading cost

Hedge Table Organisation

- For each hedge asset, record its hedge quantity, beta, variance change, score, etc. into the hedge table
- Hedge quantities are translated into BUY or SELL instructions for user interpretation
- Rank the table by hedging score in descending order
- Generate a summary text for the optimal hedging strategy

4.1. Dashboard interface

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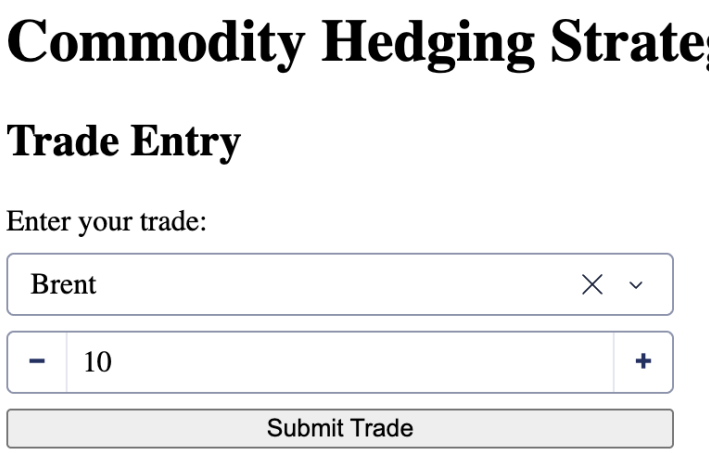


User selects the commodity and input size, then submits the exposure to trigger hedge analysis

2 The Inventory Table

Your Inventory

Product	Quantity
Brent	10
WTI	15
GO	-6
RBOB	10
HO	-8

The inventory table lists the current futures position and provides the portfolio state used in variance calculation

3 Hedge Table & Suggestion

Hedge Suggestion

Exposure +10 units of Brent. Best hedge: SELL 9.5660 units of WTI. Var before=0.067602, Var change=-0.044133, cost=0.19132, score=0.23067

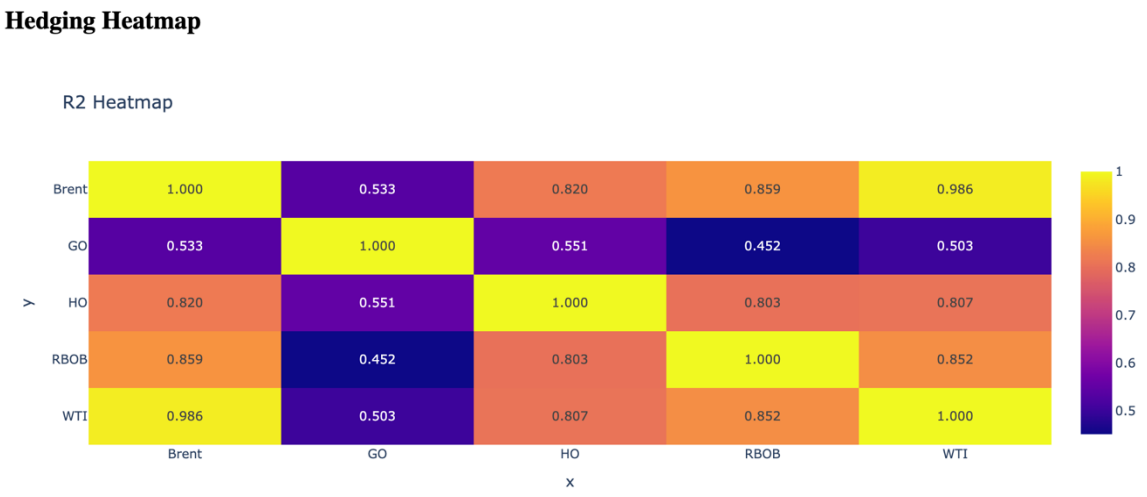
Hedge Asset	Action	Hedge Qty	Beta	R2	var_before	var_after	var_change	cost	score
WTI	SELL	9.566	0.9566	0.98586	0.0676	0.02347	-0.04413	0.19132	0.23067
RBOB	SELL	8.2232	0.82232	0.85946	0.0676	0.03041	-0.03719	0.41116	0.09045
GO	SELL	7.0701	0.70701	0.53313	0.0676	0.03792	-0.02968	0.3535	0.08396
HO	SELL	8.7634	0.87634	0.81981	0.0676	0.03281	-0.03479	0.87634	0.0397

A ranked hedge table with a concise optimal hedging suggestion that summarises asset, action, quantity , cost, and score

4.2. Dashboard interface

4 The Heatmap

- The heatmap visualizes the pairwise R^2 across future contracts
- Brighter colour indicates a better potential hedge fit between the selected exposure and hedge asset
- Clicking a heatmap cell displays the corresponding regression plot for that asset pair



5 The Regression Plot

- The regression plot shows the return relationship of asset pair selected from heatmap
- Each point represents paired return observations, while the fitted line shows the estimated linear relationship
- Helping user interpret hedge ratio β, intercept and R^2 visually